

RankGround: Efficient High-Resolution GUI Grounding via Lightweight Reranker-Guided Crop Selection

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Abstract

Graphical User Interface (GUI) grounding is a fundamental perception task for multimodal agents, enabling them to interpret natural language instructions and interact with digital interfaces. Existing methods face a fundamental trade-off between accuracy and efficiency: direct full-image inference often fails to capture small or visually similar UI elements, while multi-crop strategies improve localization at the cost of multiple expensive Vision-Language Model (VLM) calls per query.

To address this challenge, we propose **RankGround**, a two-stage framework that achieves accurate GUI grounding with a single VLM call per query. Central to our approach is **GroundRanker**, a lightweight multimodal reranker that identifies the most promising crop from a dense candidate set. Because no off-the-shelf ranking dataset is available, we construct ranking supervision data from existing grounding datasets. A strict containment criterion and boundary-aware positive augmentation improve alignment and spatial coverage in cluttered layouts. GroundRanker is then trained with a two-stage curriculum: a pointwise objective first learns coarse containment, and a listwise objective refines subtle semantic and spatial distinctions among visually similar crops. Experimental results show that RankGround consistently outperforms

strong baselines while reducing computational cost. It achieves **1.4× faster inference** and improves localization accuracy by **5.5%** on average over the second-best method across all backbones and screen scales, establishing a new state of the art in both efficiency and precision for GUI grounding.

CCS Concepts

• **Computing methodologies** → **Computer vision**; *Natural language processing*.

Keywords

GUI Grounding, Multimodal Reranking, Vision-Language Models, Region Selection, High-Resolution Screenshots, GUI Agents

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1 Introduction

GUI agents automate tasks across digital platforms [2, 21, 32]. Their success depends on *GUI grounding*, which maps a natural language instruction to a location on the screen [5]. A correct task plan still fails when the agent cannot locate the target element [31, 32]. Higher display resolutions and denser layouts make small or visually similar elements harder to distinguish [11].

Existing GUI grounding methods typically use **direct inference** or **multiple crops** [32]. **Direct inference** sends the entire screenshot through a vision-language model (VLM) once to predict target coordinates [2, 5, 24, 32]. It is fast, but small elements lose detail

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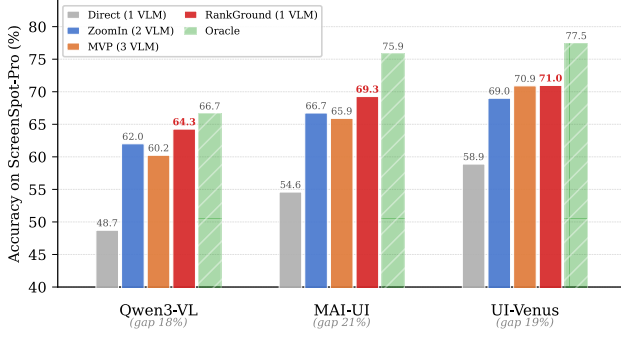


Figure 1: Accuracy of GUI grounding methods on ScreenSpot-Pro (8B scale). The gap between Direct inference and the Oracle upper bound (over 18%) reveals room for improvement. Multi-crop methods (ZoomIn, MVP) close this gap but require two to three VLM calls per query. RankGround matches or exceeds their accuracy with a single VLM call.

when a high-resolution screenshot is resized for the model. **Multi-crop** methods partition the screen and apply the VLM to each region [31, 32]. The crops expose finer detail, but the VLM now runs repeatedly. ZoomIn [32] uses two calls per query, whereas MVP [31] often uses three or more. Latency and memory use grow with the crop count, which complicates real-time deployment. Figure 1 shows the resulting cost. Similar elements in different crops can also lead to conflicting predictions.

RankGround separates region selection from coordinate prediction. Overlap tiling or sliding windows produce a dense candidate set that covers potential target locations. A lightweight multimodal reranker, **GroundRanker**, scores these crops, and the GUI-specialist VLM predicts coordinates from the top-ranked crop. The VLM therefore runs once per query while retaining local detail.

Crop selection remains difficult in dense layouts because several regions may contain similar elements. We convert instance annotations from existing grounding datasets into crop-level supervision. A crop is positive only when it fully encloses the target under a *strict containment criterion*. *Boundary-aware positive augmentation* then places the target near crop boundaries to reduce position bias.

GroundRanker follows a two-stage curriculum. Pointwise binary cross-entropy first teaches containment, and a listwise objective then separates near-miss crops from valid ones. We fine-tune Qwen3-VL-Reranker-2B [13] with *Low-Rank Adaptation (LoRA)* [6] on the query and value projections in its vision and language components. This setup learns GUI-specific spatial alignment while preserving pretrained multimodal features.

RankGround turns crop selection into a small ranking problem and reserves the costly VLM for coordinate prediction. Our main contributions follow.

- **RankGround** decouples region selection from coordinate prediction and uses one GUI-specialist VLM call per query.
- **GroundRanker** learns crop selection from strict containment labels, boundary-aware augmentation, and a pointwise-to-listwise curriculum.

- Across all tested backbones and scales, RankGround improves average localization accuracy by **5.5%** over the second-best method and runs **1.4× faster**.

2 Related Work

2.1 GUI Grounding

Early GUI grounding work adapted visual grounding methods developed for natural images [8, 12, 16, 19, 20, 27]. Some models add hierarchical low-rank adaptation [3, 26] or multimodal weight modulation [12, 28]. Such methods transfer poorly to GUI screenshots, where elements are structured and often occupy only a few pixels after resizing [2, 11]. Their scale distribution also differs sharply from natural-image datasets such as RefCOCO [30].

GUI-specific models now use dedicated data and pre-training. SeeClick [2] introduced ScreenSpot and showed the value of GUI pre-training. UGround [5], UI-TARS [21], MAI-UI [32], and UI-Venus-1.5 [24] scale this approach with more data or larger models. ScreenSpot-Pro [11] nevertheless shows that small targets on high-resolution interfaces remain difficult.

Recent methods recover visual detail through extra inference stages. ZoomIn [32] uses coarse-to-fine refinement. ScreenSeeker [11] and ZoomClick [7] repeatedly zoom into a predicted region. GOLD [9] proposes regions at low resolution and revisits them at high resolution. MVP [31] aggregates predictions from attention-guided views, whereas CoG [10] refines coordinates through self-reasoning. Training-based methods learn when or how to add these stages. GUI-ARP [29] combines supervised fine-tuning with GRPO so that the model can request another inference stage. Focus [22] learns to switch between fast prediction and a slower reasoning path. Both still spend extra forward passes on selection or verification. RankGround assigns region selection to a lightweight reranker and calls the grounding VLM once.

2.2 Multimodal Reranking

Reranking is common in information retrieval. A fast first stage narrows the candidate set before a cross-encoder scores the remaining items [18]. Recent systems extend this design to multimodal retrieval [4, 14, 25]. Qwen3-VL-Reranker [13] uses a Qwen3-VL cross-encoder to score text-image query-document pairs. RankGround applies multimodal reranking to GUI grounding and uses crop relevance scores for region selection. To our knowledge, this is the first use of a multimodal reranker for this task.

3 Method

3.1 Overview

Given a screenshot $I \in \mathbb{R}^{H \times W \times 3}$ and a natural language instruction q , GUI grounding aims to predict the normalized click coordinate $\hat{p} \in [0, 1]^2$ of the target UI element on the screenshot.

Existing multi-crop approaches ask the same VLM to select a region and predict its coordinates. The VLM runs on every candidate crop, which is costly at high resolutions.

Region selection does not require precise coordinates. It only needs to identify the crop that contains the target. We therefore treat selection as a lightweight *ranking* problem and leave one high-resolution coordinate prediction to the VLM.

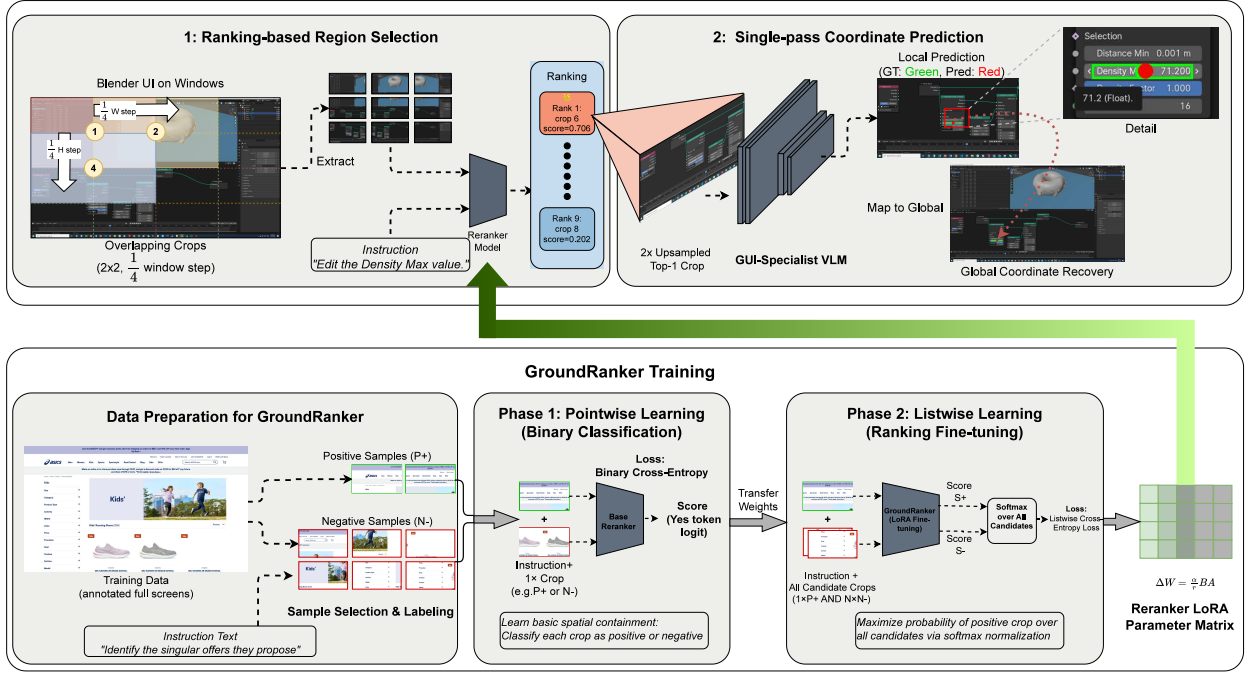


Figure 2: Overview of the RankGround pipeline. Given a screenshot and a natural language instruction, RankGround partitions the image into N crops via an overlap tiling strategy, ranks them with a fine-tuned multimodal reranker, and invokes a GUI-specialist VLM *once* on the selected crop to predict coordinates, which are mapped back to the original image space.

Let $C = \{c_1, \dots, c_N\}$ be the candidate set. Each region c_k has a top-left offset $\mathbf{o}_k$ and spatial extent $\mathbf{d}$. Prediction follows

$$c^* = \arg \max_{c_k \in C} s(q, c_k), \quad (1)$$

$$\hat{\mathbf{p}} = \mathbf{o}^* + F_\phi(q, c^*) \odot \mathbf{d}, \quad (2)$$

where $s(\cdot)$ ranks the candidate regions and F_ϕ is a GUI-specialist VLM applied once to the selected region. We instantiate s as **GroundRanker**. Section 3.2 describes how it scores crops from a coverage-preserving tiling. Section 3.3 gives the coordinate mapping for the selected crop. Figure 2 shows the full framework.

3.2 Ranking-based Region Selection

Ranking-based selection narrows the downstream VLM input to one crop. Section 3.2.1 defines candidate generation, and Section 3.2.2 derives containment labels and augmented positives. Section 3.2.3 trains GroundRanker on these crops. At inference, Section 3.2.4 passes the top crop to coordinate prediction.

3.2.1 Overlap Tiling-based Candidate Generation. A candidate must contain the full target. Otherwise, no reranker can recover it, bounding end-to-end accuracy.

Formally, let $B^* = [x_1^*, y_1^*, x_2^*, y_2^*]$ denote the ground-truth bounding box of the target element. We say that a candidate set C achieves *full coverage* if and only if

$$\exists c_k \in C \text{ s.t. } B^* \subseteq \text{box}(c_k), \quad (3)$$

where $\text{box}(c_k)$ denotes the spatial extent of crop c_k . Violating this condition makes errors irrecoverable, regardless of modeling.

We use **Overlap Tiling**, a $G \times G$ grid with overlap ratio ρ , to satisfy this condition. Each crop spans $\lfloor H/G \rfloor \times \lfloor W/G \rfloor$, and adjacent crops are shifted by $(1-\rho)$ of their extent along both axes, resulting in $N = (2G-1)^2$ candidates in total. For targets smaller than a crop, $\rho = 0.5$ guarantees Eq. (3) by applying the interval covering lemma along each axis. The crop at position (i, j) starts at

$$x_{ij} = \left\lfloor \frac{i}{2(G-1)} W \right\rfloor, \quad y_{ij} = \left\lfloor \frac{j}{2(G-1)} H \right\rfloor,$$

with crop width $w_c = \lceil W/G \rceil$ and height $h_c = \lceil H/G \rceil$, for $i, j \in \{0, \dots, 2G-2\}$. Each crop is resized to the VLM input resolution.

The size assumption matches nearly all ScreenSpot-Pro samples. Under the default $G=2$ layout, each crop covers about 25% of the screenshot and overlaps its neighbors. The resulting Oracle coverage is 99.75%. The 90th-percentile target occupies only 0.156% of the screenshot for text and 0.034% for icons. A wide target may cross every crop boundary, so Eq. (3) no longer guarantees full containment. GroundRanker can still choose a useful partial view. Figure 4 shows a ScreenSpot-V2 example in which the target spans crop boundaries. The selected crop scores 0.90, compared with 0.39 for the runner-up, and the VLM predicts the correct click.

3.2.2 Data Preparation for GroundRanker. Existing grounding datasets lack crop-level ranking labels. We derive them automatically from screenshots with ground-truth element annotations.

For a screenshot I of size $W \times H$, sliding-window tiling scales the window (w, h) to a fixed pixel budget $N_{\max}$ according to

$$r = \sqrt{\frac{N_{\max}}{W \cdot H}}, \quad w = \lfloor r \cdot W \rfloor, \quad h = \lfloor r \cdot H \rfloor. \quad (4)$$

The window moves in steps of $(w/2, h/2)$ for 50% overlap and snaps to the image boundary to avoid incomplete edge crops. Images within the pixel budget ($W \cdot H \leq N_{\max}$) use the 2×2 overlapping grid from Section 3.2.1.

Each crop c_k receives a binary **containment** label. It is positive only when it fully contains the target box $B^* = [x_1^*, y_1^*, x_2^*, y_2^*]$ under the condition

$$l_k \leq x_1^* \wedge t_k \leq y_1^* \wedge l_k + w \geq x_2^* \wedge t_k + h \geq y_2^*, \quad (5)$$

where (l_k, t_k) denotes the top-left corner of the k -th candidate crop, ensuring that the target element is entirely visible. This strict criterion is stronger than center-based or IoU-based labeling and aligns directly with the coverage condition (3).

Boundary-aware positive augmentation counters the center bias of uniform windows. Let $\Delta_x = w - (x_2^* - x_1^*)$ and $\Delta_y = h - (y_2^* - y_1^*)$ denote the slack between the crop and target box. Each positive crop yields up to 12 variants. We place the target at the four corners and four edge midpoints, then draw four more positions near the boundary. The resulting samples spread positives across the crop.

Each **listwise** [1] sample pairs one positive crop or augmented variant with every negative crop from the same image and the instruction q . This matches inference and exposes the full contrast among candidates to the loss in Eq. (9).

3.2.3 GroundRanker Training. GroundRanker is a vision-language cross-encoder that scores how likely each crop is to contain the target. It selects

$$c^* = c_{k^*}, \quad k^* = \arg \max_k s_\theta(q, c_k). \quad (6)$$

where s_θ jointly scores the crop and instruction q .

The cross-encoder processes (q, c) jointly to retain fine spatial correspondence. We use the affirmative token logit as its ranking score,

$$s_\theta(q, c) = \text{logit}_{\text{yes}}(f_\theta(q, c)), \quad (7)$$

where f_θ denotes the reranker. The score is comparable across crops and needs no classification head.

GroundRanker uses a **pointwise-to-listwise curriculum**.

Stage 1 (pointwise). Binary cross-entropy teaches spatial containment,

$$\mathcal{L}_1 = \mathbb{E}_{(q, c, y)} [-y \log \sigma(s_\theta) - (1-y) \log(1 - \sigma(s_\theta))], \quad (8)$$

where $y \in \{0, 1\}$. Boundary-aware positives reduce position bias.

Stage 2 (listwise). A listwise loss contrasts valid crops with near-miss crops that overlap the target but fail containment,

$$\mathcal{L}_2 = - \sum_{c \in C} \tilde{y}_c \cdot \log \frac{\exp(s_\theta(q, c))}{\sum_{c' \in C} \exp(s_\theta(q, c'))}, \quad (9)$$

where $\tilde{y}_c = y_c / \sum_{c'} y_{c'}$ normalizes labels. Stage 1 learns a coarse ranking. Stage 2 sharpens near-miss boundaries.

We adapt the pretrained cross-encoder with Low-Rank Adaptation (LoRA) [6]. Trainable low-rank matrices A and B update each

Table 1: Bounding-box statistics for Text and Icon elements in ScreenSpot-Pro. Rel. area is the percentage of screenshot area. P10 and P90 are the 10th and 90th percentiles.

	Text ($n=977$)			Icon ($n=604$)		
	Mean	Med.	P10/P90	Mean	Med.	P10/P90
Width (px)	141	111	36 / 272	37	26	16 / 54
Height (px)	34	25	18 / 61	29	24	17 / 48
Area (px ²)	4,655	3,186	828 / 7,680	1,764	624	305 / 2,344
Rel. area (%)	0.086	0.065	0.017 / 0.156	0.031	0.014	0.005 / 0.034

frozen weight matrix W_0 as

$$h = W_0 x + \Delta W x = W_0 x + \frac{\alpha}{r} B A x, \quad (10)$$

Here x is the input, h is the output, and ΔW is the update. The rank satisfies $r \ll \min(d, k)$, and α controls its scale. GroundRanker applies LoRA to query and value projections in the vision encoder’s self-attention and the language model’s cross-attention. This preserves pretrained features while learning spatial alignment for GUI crops.

3.2.4 Region Selection. At inference, overlap tiling or sliding windows generate C , and GroundRanker scores each crop. We select

$$c^* = \arg \max_{c_k \in C} s_\theta(q, c_k), \quad (11)$$

The selected crop then passes to coordinate prediction, so the VLM processes one region rather than the full candidate set.

3.3 Coordinate Prediction and Mapping

The VLM F_ϕ predicts a local coordinate $\hat{p}' \in [0, 1]^2$ within the upsampled crop c^* . A linear transform maps it back to the original image without approximation error.

Upsampling the crop by α yields α/G resolution relative to the full image. For $\alpha \geq G$, the VLM sees more detail per element, improving small-element localization.

Complexity analysis. Let T_R denote one batched reranker pass and T_V one VLM pass. GroundRanker scores all N crops in one batch, so region selection costs $\mathcal{O}(T_R)$ rather than $\mathcal{O}(NT_R)$. Coordinate prediction adds $\mathcal{O}(T_V)$, giving $\mathcal{O}(T_R + T_V)$ total latency. In our measurements, $T_R \approx 145$ ms and T_V ranges from about 210 to 810 ms. RankGround is slower than Direct but avoids the repeated VLM calls used by ZoomIn and MVP. Local crop processing preserves detail without the $\mathcal{O}(NT_V)$ cost of running the VLM on every candidate.

4 Experimental Setup

Dataset. We evaluate on **ScreenSpot-Pro** [11], comprising 1,581 tasks from 23 professional applications across five industries and three operating systems. We also test cross-benchmark generalization on **UI-Vision** [17]. Table 1 summarizes the bounding-box size distribution for both element types.

Evaluation Metrics. **Click Accuracy** counts predictions inside the target box and is micro-averaged over 977 **Text** and 604 **Icon/Widget** targets. We also report each type. **Reranker ablations** use **Hit Rate@k**, the fraction of queries with a valid crop among the top- k results for the corresponding grounding instruction.

Baselines. Direct inference sends the full screenshot through each backbone once. ZoomIn [32] predicts a coarse location and refines

it in a second call. MVP [31] selects attention-guided regions and aggregates several crop predictions. We test Qwen3-VL, MAI-UI, and UI-Venus-1.5 at both 8B and 2B scales. Qwen3-VL [23] uses the basic prompt, and the other models use their native prompts.

Implementation. GroundRanker is trained on 6,789 ViSurf instances. Data construction yields about 295K pairwise, 124K pointwise, and 78K listwise samples. LoRA uses rank 16, scale 32, and dropout 0.05 on query and value projections. We use AdamW with a 1×10^{-4} learning rate, 0.01 weight decay, effective batch size 8, 5% linear warmup, and cosine decay. Training converges within one epoch on RTX 4090 GPUs. Latency is wall-clock time per sample over ScreenSpot-Pro at batch size 1, including preprocessing and post-processing but excluding data loading.

5 Results and Analysis

5.1 Main Results

Table 2 shows that RankGround leads all six backbone-scale settings with one GUI-specialist VLM call and one reranker pass per query. #VLM counts GUI-specialist VLM calls, and Δ is the absolute accuracy gain (%) over Direct for the same backbone. Bold and underlining mark each scale’s best and runner-up.

RankGround improves on Direct in all six configurations, with absolute gains from 12.08 to 19.36 points. These gains require one GUI-specialist VLM call and a 145 ms reranker pass. ZoomIn uses two VLM calls. RankGround improves its average accuracy by 2.3 points at 8B and 5.4 points at 2B, and is faster for every backbone. The larger 2B gain shows that crop selection matters more when the grounder has less capacity.

RankGround also matches or exceeds MVP on every backbone. Average accuracy rises from 65.7% to 68.2% at 8B and from 50.9% to 60.2% at 2B. MVP takes 2,342–4,648 ms per sample, or 4–12 $\times$ longer, because it evaluates several crops with the VLM.

Recent best-reported ScreenSpot-Pro results provide wider context, though they use different backbones and training protocols. ScreenSeeker [11] combines GPT-4o with OS-Atlas-7B and reports 48.1% after multiple calls. GUI-ARP [29] reports 60.8% with a 7B model trained by SFT and reinforcement learning. CoG [10] reaches 68.4% with a three-call Qwen3-VL-235B/32B pipeline. RankGround reaches 64.7% at 2B and 71.0% at 8B with one grounding-VLM call plus one reranker pass. Table 2 remains the controlled comparison. These published numbers instead show where RankGround sits among systems with different model and data budgets.

Peak memory. Co-resident reranker and VLM peak at 21.2 GB for 8B and 8.6 GB for 2B, both within 24 GB. Sequential execution lowers them to 16.94 and 4.38 GB, matching Direct and staying below MVP’s 17.21 and 4.61 GB.

5.2 Ablation Studies

Section 5.2.1 examines partition geometry, Section 5.2.2 examines reranker training, and Section 5.2.3 tests prompt wording. Unless stated otherwise, the ablations use UI-Venus-1.5-2B with Overlap 2 $\times$ 2 and report mean $\pm$ std over three runs.

5.2.1 Partitioning Strategy. Table 3 compares uniform, overlapping, and sliding-window grids using the base reranker. **Oracle** assumes correct crop selection, and **Hit@1** is the fraction of valid top-ranked

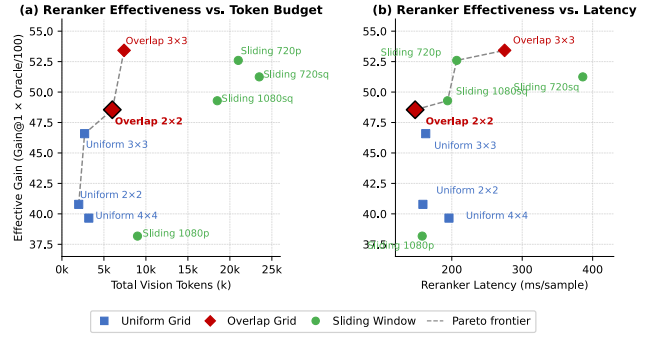


Figure 3: Effective Gain (Gain@1 $\times$ Oracle) vs. vision token budget (left) and reranker latency (right) across strategies.

crops. **Gain@1** subtracts random selection, where $\text{Rand@1} = \bar{n}_+ / N$, $\bar{n}_+$ is the mean positive-crop count, and N is the total crop count.

Oracle coverage. Finer uniform grids cover less of the screen per crop and split more boundary elements. Their Oracle accuracy falls from 93.0% at 2 $\times$ 2 to 85.9% at 4 $\times$ 4. Overlapping grids and sliding windows instead keep Oracle above 98.8%, so a 50% overlap contains nearly every target in at least one crop.

Reranker discriminability. Overlap 2 $\times$ 2 reaches 43.3 % Gain@1, or 48.7 % with the optimized prompt, while retaining 99.75 % Oracle. Sliding 1080p has the highest raw Hit@1 at 77.3 %, rising to 80.7 % with the prompt. Its random baseline is already 42.4 % because it produces only 7.2 crops on average, leaving 34.8 % Gain@1. Finer sliding windows (720p, 720sq) improve Gain@1 but degrade Hit@1 as the reranker faces more candidates with similar visual content. Overlap 3 $\times$ 3 reaches 46.8 % Gain@1, or 53.7 % with the prompt, but uses 25 crops rather than 9.

Efficiency. Overlap 2 $\times$ 2 scores 9 crops in 145 ms. Overlap 3 $\times$ 3 takes 280 ms for 25 crops, and sliding windows take up to 380 ms for 35 crops. Batched reranker latency therefore grows roughly with the crop count.

Overlap 2 $\times$ 2 reaches 99.75 % Oracle with a 9-crop search space and 43.3 % Gain@1. Its 145 ms latency is lower than larger layouts. We use it as RankGround’s default.

5.2.2 Reranker Design. Region selection quality, not VLM capacity, limits multi-crop grounding. With 2 $\times$ crops, the Oracle upper bound exceeds every multi-crop method by more than 7 points across the 8B backbones. Figure 1 shows this gap. Higher crop resolution further benefits downstream grounding even under perfect region selection, with 2 $\times$ outperforming 1 $\times$ under Oracle conditions (+3.67 % for Qwen3-VL, +1.98 % for MAI-UI, +2.91 % for UI-Venus). The base reranker remains more than 10 points below Oracle accuracy, which motivates the LoRA fine-tuning in Section 3.2. Across three seeds, the pointwise-to-listwise curriculum reaches 88.2 % Hit@1 and improves 2B Text accuracy by 0.62 points over pointwise-only training. Section 5.5 reports the full comparison.

5.2.3 Prompt Design. The chat template stores the task definition in <Instruct>, the grounding instruction in <Query>, and the crop in <Document>. Table 4 contrasts the **default prompt** inherited from Qwen3-VL-Reranker with our **optimized prompt**, which asks

Table 2: Comparison of GUI grounding methods on ScreenSpot-Pro at 8B and 2B scales.

Method	Model	#VLM	8B Scale					2B Scale				
			Time (ms)	Acc (%)	Text (%)	Icon (%)	Δ	Time (ms)	Acc (%)	Text (%)	Icon (%)	Δ
Direct	Qwen3-VL [23]	1	814±11	48.72±0.04	65.06±0.12	22.30±0.10		642±2	35.42±0.06	48.00±0.10	15.07±0.17	
	MAI-UI [32]		468±3	54.59±0.04	70.93±0.06	28.15±0.12		235±3	43.64±0.04	58.75±0.08	19.20±0.10	
	UI-Venus [24]		480±23	58.89±0.06	74.10±0.08	34.27±0.15		212±6	48.13±0.04	61.51±0.06	26.49±0.12	
ZoomIn [32]	Qwen3-VL [23]	2	1486±39	62.00±0.10	80.08±0.12	32.73±0.10	+13.28	1130±7	47.82±0.08	62.95±0.12	23.34±0.19	+12.40
	MAI-UI [32]		981±12	66.73±0.11	81.30±0.47	43.16±0.47	+12.14	505±10	57.37±0.06	73.29±0.10	31.62±0.15	+13.73
	UI-Venus [24]		1008±3	69.00±0.11	82.77±0.12	46.74±0.48	+10.11	473±3	59.20±0.08	71.24±0.10	39.74±0.17	+11.07
MVP [31]	Qwen3-VL [23]	3	4648±56	60.24±0.04	78.37±0.06	30.91±0.19	+11.52	3058±38	38.48±0.04	51.25±0.12	17.83±0.10	+3.06
	MAI-UI [32]		3500±48	65.91±0.06	80.28±0.06	42.66±0.10	+11.32	2342±32	57.39±0.04	73.40±0.10	31.70±0.12	+13.75
	UI-Venus [24]		4049±45	<u>70.90±0.04</u>	<u>83.62±0.06</u>	50.33±0.10	+12.01	2418±28	56.67±0.06	66.53±0.08	<u>40.73±0.12</u>	+8.54
RankGround	Qwen3-VL [23]	1	959±12	64.26±0.08	81.99±0.10	35.60±0.19	+15.54	787±5	52.75±0.06	69.40±0.12	25.83±0.15	+17.33
	MAI-UI [32]		613±5	69.26±0.06	82.29±0.08	48.18±0.12	+14.67	380±5	<u>63.00±0.04</u>	78.71±0.10	37.58±0.17	+19.36
	UI-Venus [24]		625±23	70.97±0.04	84.54±0.06	<u>49.01±0.10</u>	+12.08	357±7	64.71±0.06	<u>76.25±0.08</u>	46.03±0.15	+16.58

Table 3: Ablation of image partitioning strategies on ScreenSpot-Pro. Bold marks the best result per column, and underlining marks the second best. Time is average per-batch reranker latency in ms. Hit@5 is unavailable for Uniform 2×2 because it generates only four crops.

Layout	Coverage		Hit Rate@k (%) ↑			Selection Quality		Time (ms) ↓
	#Crops	Oracle (%) ↑	@1	@3	@5	Rand@1 (%)	Gain@1 (%) ↑	
Uniform 2×2 + opt. prompt	4	93.04	62.66±0.10	88.49±0.06	N/A	23.26	39.40±0.10	156±5
	4	93.04	<i>67.09±0.18</i>	<i>89.42±0.04</i>	N/A	23.26	<i>43.83±0.18</i>	159±6
Uniform 3×3 + opt. prompt	9	90.70	54.97±0.17	74.26±0.17	83.24±0.11	10.08	44.89±0.17	160±6
	9	90.70	<i>61.44±0.26</i>	<i>79.38±0.06</i>	<i>86.32±0.07</i>	10.08	<i>51.36±0.26</i>	163±5
Uniform 4×4 + opt. prompt	16	85.90	45.86±0.29	62.98±0.42	70.15±0.19	5.37	40.49±0.29	200±8
	16	85.90	<i>51.53±0.46</i>	<i>67.95±0.35</i>	<i>75.35±0.07</i>	5.37	<i>46.16±0.46</i>	196±7
Overlap 2×2 + opt. prompt	9	99.75	67.13±0.13	88.32±0.10	94.73±0.04	23.84	43.29±0.13	145±4
	9	99.75	<i>72.51±0.35</i>	<i>91.21±0.17</i>	<i>96.16±0.04</i>	23.84	<i>48.67±0.35</i>	148±5
Overlap 3×3 + opt. prompt	25	99.43	56.27±0.42	75.29±0.13	83.49±0.33	9.50	46.78±0.42	280±12
	25	99.43	<i>63.23±0.19</i>	<i>81.07±0.44</i>	<i>87.92±0.46</i>	9.50	<i>53.73±0.19</i>	275±11
Sliding 1080p + opt. prompt	~7.2	99.75	77.25±0.04	93.28±0.13	92.86±0.15	42.44	34.81±0.04	155±6
	~7.2	99.75	<i>80.71±0.33</i>	<i>93.64±0.04</i>	<i>93.69±0.13</i>	42.44	<i>38.27±0.33</i>	158±7
Sliding 1080sq + opt. prompt	~14.8	99.43	65.72±0.17	84.06±0.17	91.08±0.16	21.96	43.75±0.17	190±8
	~14.8	99.43	<i>71.54±0.33</i>	<i>87.73±0.25</i>	<i>93.21±0.17</i>	21.96	<i>49.57±0.33</i>	194±9
Sliding 720p + opt. prompt	~17.7	99.62	64.54±0.26	81.72±0.11	89.05±0.13	17.79	46.75±0.26	210±9
	~17.7	99.62	<i>70.59±0.32</i>	<i>86.27±0.13</i>	<i>91.49±0.17</i>	17.79	<i>52.80±0.32</i>	207±8
Sliding 720sq + opt. prompt	~35.2	98.80	54.33±0.35	73.77±0.29	80.83±0.22	8.70	45.63±0.35	380±15
	~35.2	98.80	<i>60.57±0.10</i>	<i>78.37±0.29</i>	<i>84.69±0.23</i>	8.70	<i>51.87±0.10</i>	386±14
LoRA FT reranker, Overlap 2×2 + opt. prompt	9	99.75	<u>85.52±0.19</u>	<u>93.36±0.11</u>	<u>97.41±0.08</u>	23.84	<u>61.68±0.19</u>	145±4
	9	99.75	88.17±0.14	94.69±0.08	97.72±0.06	23.84	64.33±0.14	147±5

directly whether the crop contains the target. The optimized prompt consistently improves Hit@1 over the default prompt across all partitioning strategies. The largest gain is +5.4 points on Overlap 2×2, shown by the italicized rows of Table 3. The prompt gives the reranker a clearer containment question without changing its weights.

Table 4: Default and optimized reranker prompts.

Default	<i>“Given a search query, retrieve relevant candidates that answer the query.”</i>
Optimized	<i>“Given a GUI operation instruction, determine whether the provided image region contains the target UI element described in the instruction.”</i>

5.3 Reranker Generalization

Table 5 shows 78.70 % Hit@1 on UI-Vision [17] without benchmark-specific fine-tuning.

Table 5: Hit Rate (HR) on out-of-distribution benchmarks using Overlap 2×2 and UI-Venus-1.5-2B.

Method	Benchmark	HR@1 (%)	HR@3 (%)	HR@5 (%)
Base reranker	UI-Vision	52.93±0.11	79.02±0.09	90.16±0.08
RankGround	UI-Vision	78.70±0.00	89.05±0.00	95.26±0.00

Table 6 reports 49.00 % end-to-end accuracy for RankGround on UI-Vision, compared with 18.88 % for Direct.

Table 6: Oracle coverage and Accuracy on out-of-distribution benchmarks using Overlap 2×2 and Qwen3-VL-2B.

Method	Benchmark	Oracle (%)	Acc (%)
Direct	UI-Vision	N/A	18.88±0.06
Base reranker	UI-Vision	98.36	25.76±0.10
RankGround	UI-Vision	98.36	49.00±0.06

The reranker is trained only on ViSurf [15], an OmniACT-derived dataset with 79% desktop and 21% web data from 62 applications. ScreenSpot-Pro and UI-Vision therefore differ from its training distribution. Table 7 shows that UI-Vision also differs from ScreenSpot-Pro. It contains 2,786 Icon samples and 758 Text samples. Element sizes are similar in pixels, but their relative areas are about 2× larger, which indicates lower-resolution screenshots on average.

Table 7: Bounding-box statistics for Text and Icon elements in UI-Vision. Rel. area is the percentage of screenshot area. P10 and P90 are the 10th and 90th percentiles.

	Text ($n=758$)			Icon ($n=2,786$)		
	Mean	Med.	P10/P90	Mean	Med.	P10/P90
Width (px)	136	108	38 / 275	41	34	22 / 58
Height (px)	32	26	19 / 51	32	28	20 / 46
Area (px ²)	4,498	3,002	870 / 10,135	1,435	962	459 / 2,592
Rel. area (%)	0.209	0.161	0.045 / 0.454	0.069	0.052	0.027 / 0.112

5.4 Broader Benchmarks

We further test UI-Venus-1.5-8B on five benchmarks that cover web, desktop, and operating-system interfaces. Table 8 uses the same grounder for Direct, ZoomIn, and RankGround within every benchmark.

Table 8: Click accuracy (%) on five additional benchmarks with UI-Venus-1.5-8B. Δ is the gain over ZoomIn.

Benchmark	Direct	ZoomIn	RankGround	Δ
ScreenSpot-V2	96.31	94.18	95.36	+1.18
MMBench-GUI-L2	87.48	85.84	89.84	+4.00
OSWorld-G	74.71	72.55	78.04	+5.49
OSWorld-G-Refined	80.20	80.59	84.71	+4.12
VenusBench-GD	84.71	85.54	89.62	+4.08

RankGround beats ZoomIn on all five benchmarks by 1.18–5.49 points. It also exceeds Direct by 2.36–4.91 points on MMBench-GUI-L2, OSWorld-G, OSWorld-G-Refined, and VenusBench-GD. ScreenSpot-V2 is the one exception: Direct is already at 96.31%, while RankGround stays within one point and remains above ZoomIn. The gains on the other four sets show that learned crop selection transfers beyond ScreenSpot-Pro and UI-Vision.

5.5 Curriculum Analysis

We compare the two-stage curriculum with Pointwise-only and Listwise-only training. The data, LoRA configuration, optimizer, and evaluation protocol remain fixed. Each result is averaged over

three random seeds. Pointwise-only uses $\mathcal{L}_1$ throughout, while Listwise-only starts directly from $\mathcal{L}_2$ without containment pre-training.

Table 9: Reranker accuracy (%) for each training objective on ScreenSpot-Pro. Results are mean±std over three runs.

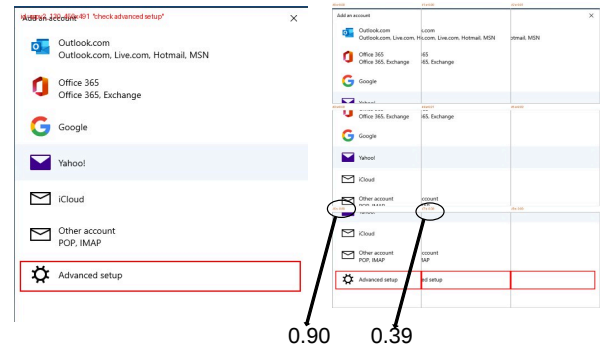
Objective	Hit@1	Hit@3	Hit@5	Text@1	Icon@1
Pointwise-only	87.86±0.18	94.24±0.11	97.91±0.07	92.84±0.22	79.80±0.31
Listwise-only	88.17±0.16	94.75±0.10	97.98±0.06	93.86±0.19	78.97±0.34
Two-Stage	88.20±0.15	94.82±0.09	98.05±0.06	94.28±0.18	78.36±0.30

Table 9 shows that the curriculum gives the best Hit@1, Hit@3, Hit@5, and Text Hit@1. Pointwise-only remains strongest on icons. Small icons often need only a binary containment decision, while the listwise stage helps more when several text-bearing crops are semantically similar. The curriculum also reaches 90.00% validation AUROC and completes its listwise stage at step 260, 30 steps earlier than Listwise-only.

Table 10: End-to-end accuracy (%) for each reranker objective on ScreenSpot-Pro. Results are mean±std over three runs.

Objective	Model	Acc	Text	Icon
Pointwise-only	Qwen3-VL-2B	52.62±0.32	68.58±0.41	26.82±0.58
Listwise-only	Qwen3-VL-2B	52.75±0.30	68.78±0.39	26.82±0.56
Two-Stage	Qwen3-VL-2B	52.90±0.28	69.20±0.35	26.53±0.54
Pointwise-only	Qwen3-VL-8B	65.72±0.28	83.01±0.33	37.75±0.51
Listwise-only	Qwen3-VL-8B	65.34±0.30	82.50±0.35	37.58±0.53
Two-Stage	Qwen3-VL-8B	65.85±0.26	83.20±0.31	37.78±0.49

The ranking gains carry into coordinate prediction. Two-Stage gives the highest overall and Text accuracy at both model scales. At 2B, it improves overall accuracy by 0.28 points over Pointwise-only and Text accuracy by 0.62 points. The smaller gains at 8B suggest that a stronger coordinate predictor can absorb some crop-selection noise, but it still benefits from the curriculum.

**Figure 4: Wide-target example on ScreenSpot-V2. The target crosses crop boundaries, yet GroundRanker selects the useful crop with a clear score margin.**

5.6 Failure Modes and Fallbacks

Overlap Tiling covers the target in 99.75% of ScreenSpot-Pro samples, while GroundRanker reaches 88.2% Hit@1. Coverage failures therefore account for only 0.25% of the data. Most first-stage errors come from choosing the wrong candidate among valid crops. They concentrate on pure icons: the reranker error rate is 19.9% for icons and 5.5% for text. Repeated visual motifs and weak textual cues make icons harder to distinguish. Without the target in the selected crop, the coordinate predictor cannot recover.

The errors also cluster by application. Table 11 lists the ten hardest applications in ScreenSpot-Pro. Origin has a 41.9% error rate, almost twice the 21.0% rate of Windows Common. Its toolbars contain many chart-type icons with similar shapes, and instructions such as “plot candlestick chart” provide little local text for matching.

Table 11: Applications with the highest GroundRanker error rates on ScreenSpot-Pro.

Application	OS	Total	Errors	Err. (%)
Origin	Win	62	26	41.9
Windows Common	Win	81	17	21.0
Android Studio	mac	80	14	17.5
DaVinci Resolve	mac	44	7	15.9
macOS Common	mac	65	10	15.4
Premiere Pro	Win	52	8	15.4
Fruitloops	Win	57	8	14.0
PyCharm	mac	78	9	11.5
Vivado	Win	80	9	11.2
MATLAB	mac	93	9	9.7

The score margin gives a second view of these errors. Among incorrect selections, 54.6% have a Top-1 minus Top-2 margin below 0.05, while only 7.5% exceed 0.2. Most errors are therefore close decisions between plausible crops rather than confident mismatches. Figure 5 shows this concentration near zero.

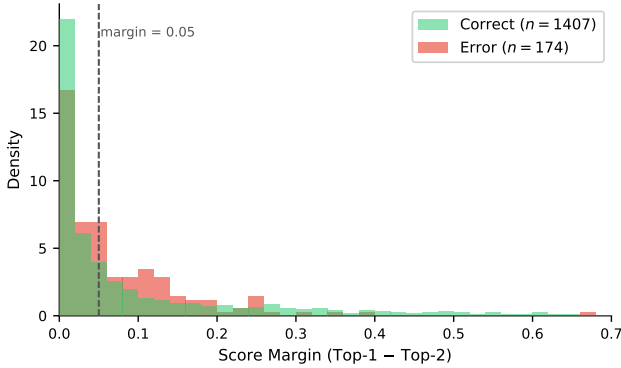


Figure 5: Score margins for correct and incorrect crop selections. The dashed line marks a margin of 0.05, below which 54.6% of errors fall.

Figure 6 shows the two main failure patterns. In the Blender case, several crops contain plausible matches and the top two scores differ by only 0.001. In the Origin case, the chart icons remain hard to tell apart at crop scale, and every score stays below 0.42.

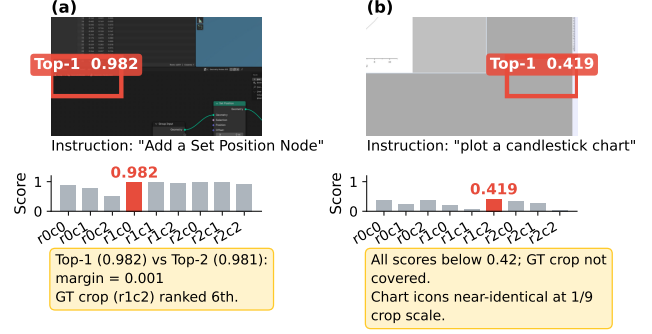


Figure 6: Representative failures on ScreenSpot-Pro. The red box marks the selected crop. Panel (a) is a near tie, while panel (b) contains visually similar chart icons.

We also test whether extra crops can rescue these cases. For Top- k , the grounding VLM processes each of the highest-ranked crops, then coordinate-token confidence chooses the final prediction. Top-1 gives 71.0% accuracy with 1.0 VLM call per query. Top-2 gives 64.0% with 2.0 calls, and Top-3 gives 56.9% with 3.0 calls. A confidence-triggered fallback reaches 68.4% with 1.47 calls on average. Coordinate-token confidence does not compare predictions across crops as reliably as GroundRanker compares the crops themselves. We therefore retain Top-1, which gives the best measured accuracy and the lowest cost.

6 Limitations

RankGround depends on one selected crop. Its 99.75% Oracle coverage assumes that the target is smaller than a crop, so wide controls such as Figure 4 have no formal containment guarantee. Hit@1 is 88.2%, and our Top- k experiments show that coordinate-token confidence cannot reliably recover a wrong selection. A better fallback would need confidence calibrated across crops.

GroundRanker was trained on ViSurf, whose data are 79% desktop and 21% web. Transfer to UI-Vision and five other benchmarks is encouraging, but mobile-heavy, multilingual, and changing interfaces remain untested. Pure icons also have a 19.9% selection error rate versus 5.5% for text. The fixed 145 ms reranker pass is modest beside an 8B VLM but more visible at 2B. RankGround is faster than repeated-call baselines, not Direct, and offers an accuracy-efficiency trade-off.

7 Conclusion

On ScreenSpot-Pro, the gap between Direct and Oracle exceeds 18 points. Region selection, rather than coordinate prediction, accounts for most of this gap. An Oracle selector with one VLM call also beats ZoomIn and MVP by more than 7 points across the 8B backbones.

RankGround replaces repeated VLM-based selection with GroundRanker. Strict containment labels and boundary-aware augmentation teach it which crops preserve a target, while the pointwise-to-listwise curriculum separates near misses. Across backbones and scales, it gains 5.5% over the best multi-crop baseline and runs 1.4× faster.

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